

Physics Extended Essay – Detailed Structure Plan (v3)

Investigation on the frequencies of oscillations of a coupled
magnetic–mechanical oscillator

Working document: Edwin’s skeleton + INTERNATIONAL’s evaluation loop

September 8, 2026

Storyline (keep this on your wall)

Two identical copper leaf springs with repelling magnets form two coupled oscillators. The in-phase mode (f_1) should not feel the magnets; the anti-phase mode (f_2) is stiffened by the magnetic coupling γ . Modelling the magnets as point dipoles gives

$$f_2^2 - f_1^2 = \frac{3\mu_0\mu_m^2}{\pi^3 m d^5},$$

a d^{-5} power law. f_1, f_2 are measured by video tracking + FFT for $d = 21\text{--}60\text{ mm}$. The $\ln\text{--}\ln$ gradient was -3.75 ± 0.19 , f_1 drifted by 2.6 %, and the peaks could not be separated beyond 35 mm. The evaluation separates *measurement* errors (FFT resolution, leakage, bin centring, damping linewidth) from *model* errors (amplitude non-linearity of the r^{-4} force – a *hardening* effect; finite magnet size – a *softening* effect; non-identical springs; tilt), each with sign, size and remedy. A modified method (normal-coordinate tracking, longer records, small release, time-domain two-tone fitting, non-linear least squares) plus an RK4 simulation of the full non-linear equations quantifies each effect and recovers the dipole law within its range of validity ($d \gtrsim 4D$).

Corrections relative to the earlier drafts (v1 and v2)

- Amplitude non-linearity of the r^{-4} force is a **hardening** effect: f_2 is biased *upward* at small d and the $\ln\text{--}\ln$ gradient becomes *steeper* ($-5.00 \rightarrow -5.16$). It cannot explain the measured -3.75 . The *softening* that flattens the gradient is the **finite-size** effect. (v2 had these merged, with the wrong sign.)
- Detuned springs: only ω_1 is pushed toward $\bar{\omega}$; ω_2 grows without bound. (v2: “both modes are pushed towards $\bar{\omega}$ ” – wrong.)
- IYPT 2023 problem 6 is “Magnetic-Mechanical Oscillator” (not “Magnetic Pendulums”).
- $\Delta d = 0.5\text{ mm}$ at $d = 21\text{ mm}$ is 2.4 % on d but $\approx 12\%$ on $Y \propto d^{-5}$.
- Damping linewidth of an *amplitude* spectrum (what Tracker/Capstone show) is $\sqrt{3}/(\pi\tau) \approx 0.022\text{ Hz}$ at $\tau = 25\text{ s}$, not $1/(\pi\tau)$ (that is the power spectrum).
- Plucking a spring gives k/m only; the absolute test of C still needs m , hence a static k . Pluck each spring *with its own magnet mounted*, other magnet removed.
- Assessment criteria: from May 2027 the EE is marked out of 30 (A 6, B 6, C 6, D 8, E 4); the “Critical thinking 12 / Presentation / Engagement” scheme is the legacy 34-mark model. Confirm your session with your supervisor.
- Word budget now sums to 3950 (v1 summed to 4250).

Contents

1 Research question and definitions

2

2	Structure map (Edwin → yours → INTERNATIONAL)	2
3	Section-by-section content	3
4	FFT error analysis – what to write	7
5	“Non-linearisation” – three levels, use all three in order	8
6	Systematic-error summary table (Table 7 template)	8
7	Lab / data checklist	9
8	Assessment-criteria map (first assessment May 2027, 30 marks)	9
9	Figures and tables (final list for the essay)	9
10	Supporting figures already generated	9
11	Appendices to write	13
12	Ready-made numbers	13
13	Pitfalls (examiner’s eye)	13

1 Research question and definitions

RQ. *How does the equilibrium centre-to-centre separation d of the magnets in a coupled magnetic-mechanical oscillator affect its two normal-mode frequencies f_1 and f_2 , and does the coupling term $f_2^2 - f_1^2$ follow the inverse-fifth-power law predicted by the magnetic-dipole model?*

It names the independent variable (d), the dependent variables (f_1, f_2), the quantity actually tested ($f_2^2 - f_1^2$) and the theoretical claim (d^{-5}). Keep the title; put the RQ in §1.3.

Definition of d (state once in §4, never mix conventions). d is the centre-to-centre separation of the two magnets at equilibrium with both magnets mounted. For identical magnets of thickness t and face-to-face gap g : $d = g + t$. Read g from a video frame that contains a scale bar (calipers between repelling magnets on flexible springs disturb the equilibrium).

Symbols. x_1, x_2 : tip displacements, positive in the same laboratory direction; m : effective oscillator mass (magnet + part of the spring); M : magnet mass; μ : linear density of the spring; μ_m : magnetic dipole moment; k : tip stiffness; γ : magnetic coupling stiffness; $Y \equiv f_2^2 - f_1^2$; D, t : magnet diameter and thickness; T : record length; f_s : frame rate.

2 Structure map (Edwin → yours → INTERNATIONAL)

Budget ≈ 3950 words for a 4000-word essay (equations, tables, captions, appendices, references are not counted). If you overrun, trim §6.1 and §9 first; protect §10–13.

#	Edwin	Yours (section title)	Words	From INTERNATIONAL
1	Introduction 1.1–1.3	1.1 The oscillator (done) · 1.2 Modes and beats · 1.3 Motivation & RQ	350	motivation paragraph (their §1.4)
2	Qualitative prediction	2 Qualitative prediction	70	
3	Hypothesis	3 Hypothesis (quantitative: gradient –5)	50	
4	Variable table	4 Variables (table; definition of d ; FFT settings as controlled)	60	FFT settings as controlled variables
9	Theoretical model (late in Edwin)	5 Theoretical model (moved <i>before</i> methodology)	500	5.4 remarks: units, assumptions
5–7	Methodology, data collection, procedures	6 Methodology: 6.1 FFT background · 6.2 set-up · 6.3 data collection (attempt 1 → 2) · 6.4 procedures	600	6.1 mirrors their §6.1
8	Data	7 Raw data (+ “resolved?” column)	180	half-range justification
10	Processed data	8 Analysis: table · ln–ln graph · Y vs d^{-5} · f_1 vs d · comparison	420	
11	Discussion & conclusion	9 Discussion (preliminary)	230	
12	Evaluation	10 Evaluation: 10.1 measurement · 10.2–10.5 model · 10.6 random	500	sign, size, remedy for each
–	–	11 Further methodology	200	their §11
–	–	12 Further data & discussion	320	their §12–13
–	–	13 Numerical simulation (RK4)	250	their §14
11.2, 13	Conclusion, extension	14 Conclusion & extensions	220	
A–E	Appendices	A–H (Section 11)	0	

Key equations for the essay

dipole force (like poles facing):
$$F(r) = \frac{3\mu_0\mu_m^2}{2\pi r^4}, \quad \gamma = -F'(d) = \frac{6\mu_0\mu_m^2}{\pi d^5} \quad (1)$$

normal modes:
$$\omega_1^2 = \frac{k}{m}, \quad \omega_2^2 = \frac{k + 2\gamma}{m} \quad (2)$$

tested relation:
$$Y = f_2^2 - f_1^2 = \frac{\gamma}{2\pi^2 m} = \frac{3\mu_0\mu_m^2}{\pi^3 m d^5} \equiv C d^{-5} \quad (3)$$

beats:
$$4f_{\text{avg}}f_{\text{beat}} = f_2^2 - f_1^2 \quad (4)$$

uncertainties:
$$\Delta Y = 2(f_1\Delta f_1 + f_2\Delta f_2), \quad \Delta(\ln Y) = \frac{\Delta Y}{Y} \quad (5)$$

$$\Delta(\ln d) = \frac{\Delta d}{d}, \quad \frac{\Delta(d^{-5})}{d^{-5}} = 5 \frac{\Delta d}{d}$$

3 Section-by-section content

1 Introduction (≈350 words)

1.2 Modes of oscillation and beats. Translational displacements x_1, x_2 of the magnets. In-phase mode (f_1): magnets move together, separation constant, magnets “invisible”. Anti-phase mode (f_2): separation oscillates, magnetic force does work. Any release is a superposition $\Rightarrow$ beats with $f_{\text{avg}} = (f_1 + f_2)/2$, $f_{\text{beat}} = (f_2 - f_1)/2$; include identity (4) – it links your test quantity to Edwin’s average/beat language and explains why beats vanish at large d . Fig. 2: your own sketch of the modes. Fig. 3: one of your own $x(t)$ traces showing beats.

1.3 Motivation & RQ. Origin: IYPT 2023, problem 6 “Magnetic-Mechanical Oscillator” (cite the official problem set). Analogy: contactless (magnetically) coupled resonators. Personal angle: the coupling can be tuned continuously by moving the springs, unlike a fixed spring. State the RQ and the definition of d .

2 Qualitative prediction 3 Hypothesis (≈120 words)

Repulsion $\propto d^{-4}$, so the extra stiffness of the anti-phase mode $\gamma = -dF/dd \propto d^{-5}$ collapses quickly with d : $f_2 \rightarrow f_1$ and the beat period $\rightarrow \infty$; f_1 is unchanged.

Hypothesis: f_1 independent of d ; $f_2^2 - f_1^2 \propto d^{-5}$, i.e. $\ln(f_2^2 - f_1^2)$ against $\ln d$ is a straight line of gradient -5 .

4 Variables (≈60 words)

Independent d (centre-to-centre = $g + t$, magnets mounted, at equilibrium).

Dependent f_1, f_2 (hence $Y = f_2^2 - f_1^2$).

Controlled same spring pair; clamped length L ; clamp torque; same magnet pair and orientation; initial displacement (size, which spring); release method; camera fps, resolution, distance; record length T ; FFT settings (window, N , zero-padding, detrend).

Uncontrolled air currents; temperature (E of copper); residual clamp slip.

5 Theoretical model (≈500 words)

5.1 Leaf spring as an oscillator. Cantilever tip stiffness $k = 3EI/L^3$ and effective mass $m = M + \frac{33}{140}\mu L$ (Rayleigh) give $\omega_0^2 = k/m$ (Appendix A). One sentence: the Euler–Bernoulli continuum model of the IYPT slides reduces to this for a heavy tip mass; an FEM check shows $< 1\%$ difference in Y for $M/(\mu L) \geq 0.4$. *Calibration strategy:* (i) the *ratio* k/m for each spring comes from plucking it with its own magnet mounted and the other magnet removed (f_a, f_b ; this bypasses E , the Rayleigh factor and the lumped approximation, and directly gives the detuning of §10.4); (ii) the *absolute* test of C in eq. (3) additionally needs m , obtained as $m = k/\omega_a^2$ from the static k (load–deflection) – so the static measurement is required, not a mere cross-check. The gradient test (-5) needs neither.

5.2 Magnetic force. Coaxial identical dipoles, like poles facing, eq. (1) (from $\mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B})$, Appendix B). Equilibrium: springs bend outward by δ with $k\delta = F(d)$. Because d is measured directly, the IYPT quintic for the equilibrium point is not needed – say so.

5.3 Linearised coupling and normal modes. $F(d + x_2 - x_1) \approx F(d) - \gamma(x_2 - x_1)$; the constant $F(d)$ only sets the equilibrium and is dropped, so

$$m\ddot{x}_1 = -kx_1 + \gamma(x_2 - x_1), \quad m\ddot{x}_2 = -kx_2 - \gamma(x_2 - x_1). \quad (6)$$

Normal coordinates $\eta = x_1 + x_2$, $\xi = x_1 - x_2$ (matrix form like Edwin’s eqs. 12–15) give eq. (2) and the boxed result (3). Graph forms: (i) $\ln Y = \ln C - 5 \ln d$; (ii) $Y = C d^{-5}$, a line through the origin whose gradient is compared with C from independently measured μ_m and m .

5.4 Remarks (INTERNATIONAL §5.3 style). Units: $[\mu_0 \mu_m^2 / (m d^5)] = 1/\text{s}^2$. Assumptions to revisit: (a) point dipoles; (b) linearised force (small amplitude); (c) identical springs; (d) lumped mass; (e) no damping; (f) magnets coaxial (no tilt); (g) the FFT returns the true frequencies.

6 Methodology (≈600 words)

6.1 FFT background – INTERNATIONAL §6.1 adapted to *frequency* measurement (content in Section 4).

6.2 Set-up & apparatus: labelled photo (Edwin Fig. 3); apparatus list; magnets (grade, $D \times t$, M); springs (L, b, h, μ); camera (model, fps, resolution); tracking software.

6.3 Method of data collection – attempt 1 (beat and average periods read from $x(t)$) fails at large d because the envelope disappears when $f_2 - f_1$ is small; attempt 2 (FFT of $x(t)$) adopted.

Show one $x(t)$ and its FFT with the two peaks labelled (Edwin Figs. 4–6).

6.4 Procedures

6.4.1 static stiffness k : load–deflection, F vs x , LINEST (Edwin §7.1);

6.4.2 dynamic calibration: f_a, f_b of each spring with its own magnet, other magnet removed (pluck, FFT, ≥ 60 s);

6.4.3 magnet moment μ_m : one magnet on an electronic balance, the other coaxial above it at measured r , $F = \Delta m g$ for 4–6 values of r , fit $F \propto r^{-4}$ (independent check of the force law and of μ_m ; datasheet B_r as a cross-check, $\mu_m = B_r V / \mu_0$);

6.4.4 masses M, μ (weigh a known length of spring);

6.4.5 f_1, f_2 vs d : set d , displace one spring by A , release, record T s at f_s , track, resample to uniform time, detrend, FFT, read the two highest peaks; 3–5 trials per d .

7 Raw data (≈ 180 words)

Table 1 constants ($L, b, h, M, \mu, k \pm \Delta k, \mu_m \pm \Delta \mu_m, m, f_a, f_b$). Table 2 F vs x . Tables 3/4 f_1, f_2 per trial with mean and half-range (justify half-range vs SD as INTERNATIONAL §8.2: $n = 3$ –5 is too small for SD). **Caption must state T, f_s, N and the bin width $1/T$** ; add a column “resolved? ($f_2 - f_1 \geq 2/T$)”. Rows $d = 40$ –60 mm: “peaks unresolved” – a result, not a gap.

8 Analysis (≈ 420 words)

8.1 Processed table: $d, \ln d \pm \Delta d/d, Y \pm \Delta Y, \ln Y \pm \Delta Y/Y, d^{-5} \pm 5(\Delta d/d)d^{-5}$ [eq. (5)]; one sample calculation.

8.2 Graph 1: $\ln Y$ vs $\ln d$ with error bars on *both* axes ($\Delta(\ln d) = \Delta d/d \approx 2$ –5 % is comparable to $\Delta(\ln Y)$ at small d), best/worst lines or LINEST. Report the unweighted gradient -3.75 ± 0.19 and the weighted one -3.48 ± 0.24 ; the difference itself exposes the $\ln$ -transform problem (Fig. 1).

8.3 Graph 2: Y vs d^{-5} (theory: line through the origin, gradient C). Graph 3: f_1 vs d (theory: horizontal) – the 2.6 % drift will be visible.

8.4 Comparison table (Edwin Tables 13/14): gradient -5 vs measured; C from μ_m, m vs gradient of Graph 2; percentage differences.

9 Discussion – preliminary (≈ 230 words)

Trend confirmed (Y falls steeply, $f_2 \rightarrow f_1$), but: exponent $-3.75 \neq -5$; f_1 not constant; no separable peaks beyond 35 mm; residuals of the $\ln$ – $\ln$ fit are systematic (concave-down: $-, +, -, +, +, -$), not random. Announce the two families of causes – *measurement* (§10.1) and *model* (§10.2–10.5) – and hand over.

10 Evaluation (≈ 500 words)

One sub-section per effect; each ends with **sign, size, remedy**. Table 7 summarises them (template in Section 6).

10.1 FFT resolution, leakage and linewidth (measurement). Two tones closer than $\sim 1/T$ (rectangular) or $\sim 2/T$ (Hann) merge or pull each other; $f_2 - f_1 = 0.039$ Hz at 35 mm needs $T \geq 26$ –51 s, and 0.009 Hz at 40 mm needs 110–220 s (table in Section 4; Fig. 2). The damping linewidth (≈ 0.02 Hz for $\tau \approx 25$ s) is a floor that no record length removes. **Sign:** Y biased at large d (both directions possible; unresolved points lost). **Size:** ± 2 –10 mHz peak pulling at 21–35 mm for $T = 20$ s; total loss at 40 mm. **Remedy:** longer T , normal-coordinate tracking, two-tone fit (§11).

10.2 Amplitude non-linearity of the magnetic force (model, assumption b). The r^{-4} force is convex, so a finite anti-phase swing samples a *stiffer* average force than the tangent stiffness γ – a *hardening* non-linearity. Expanding $F(d - \xi)$ (Appendix D):

$$\ddot{\xi} + \omega_2^2 \xi + \alpha \xi^2 + \beta \xi^3 = 0, \quad \alpha = \frac{5}{2} \frac{g \omega_2^2}{d}, \quad \beta = \frac{5}{d^2} g \omega_2^2, \quad g \equiv \frac{f_2^2 - f_1^2}{f_2^2}, \quad (7)$$

and the Landau–Lifshitz frequency shift

$$\frac{\Delta\omega}{\omega_2} = \left(\frac{a}{d}\right)^2 \left(\frac{15g}{8} - \frac{125g^2}{48}\right) > 0, \quad a = \text{amplitude of } \xi. \quad (8)$$

Sign: f_2 overestimated, most at small d ; the ln–ln gradient becomes *steeper* ($-5.00 \rightarrow -5.16$ in the RK4 simulation) – this effect cannot explain the measured -3.75 , and must be corrected before the softening effects are assessed. **Size:** $A = 10$ mm release: $+7.6\%$ at 21 mm (perturbation), $+9.8\%$ (RK4); $+1.2\%$ at 35 mm. $A = 2$ mm: $+0.3\%$ and $+0.03\%$ (Figs. 3, 4). **Remedy:** $A \leq 3$ mm; analyse late (small-amplitude) segments; extrapolate f_2 to $A \rightarrow 0$; RK4 correction (§13). *Tilt sub-effect* (assumption f): cantilever tip slope $\theta \simeq 3A/(2L)$ (0.10 rad = 5.7° for $A = 10$ mm, $L = 150$ mm); the dipole–dipole energy $\propto 3\cos\theta_1\cos\theta_2 - \cos(\theta_1 - \theta_2)$ is weakened by $\approx 0.5\%$ (mirror-symmetric tilt) to 1.5% (same-direction tilt) – negligible; one sentence and the number.

10.3 Finite magnet size (model, assumption a). The dipole law is the $d \gg D$ limit; here $d/D \approx 2$ –6. Replacing each magnet by two uniformly “charged” discs (Gilbert model, Appendix E) gives a force that falls *more slowly* than d^{-4} at short range – a *softening* of the law. **Sign:** apparent gradient flatter than -5 . **Size:** over 21–35 mm: -4.9 ($D = 6$ mm), -4.5 (12 mm), -3.75 (20 mm) (Fig. 5; rerun `finite_size_magnets.py` with your D, t). **Remedy:** use the finite-size force in the model, or larger d – which collides with 10.1; discuss the trade-off explicitly.

10.4 Non-identical springs (model, assumption c; Appendix C). With uncoupled frequencies $\omega_a \neq \omega_b$,

$$\omega_{1,2}^2 = \bar{\omega}^2 + \kappa \mp \sqrt{\Delta^2 + \kappa^2}, \quad \kappa = \frac{\gamma}{m}, \quad \Delta = \frac{\omega_b^2 - \omega_a^2}{2}, \quad \bar{\omega}^2 = \frac{\omega_a^2 + \omega_b^2}{2}, \quad (9)$$

$$\text{hence} \quad Y^2 = (f_b^2 - f_a^2)^2 + (Cd^{-5})^2. \quad (10)$$

Behaviour: as κ grows, ω_1 rises from ω_a toward $\bar{\omega}$ (and no further), while ω_2 increases without bound. **Sign:** f_1 rises as d shrinks (observed $0.802 \rightarrow 0.823$ Hz; the model with $f_a = 0.793$, $f_b = 0.849$ Hz gives $0.798 \rightarrow 0.819$ Hz, Fig. 6); Y saturates at $|f_b^2 - f_a^2|$ instead of $\rightarrow 0$, flattening the large- d end. **Size:** set by the measured f_a, f_b (§6.4.2). **Remedy:** fit the detuned formula with f_a, f_b fixed from the pluck measurement.

10.5 Damping and minor effects. Frequency shift from damping $\approx \frac{1}{2}(1/4\pi f\tau)^2$ – negligible for $\tau \sim 20$ –30 s; its linewidth matters for 10.1. Lumped-mass assumption (d): $< 1\%$ (Appendix A). Distance uncertainty: $\Delta d = 0.5$ mm at 21 mm is 2.4% on d but $\approx 12\%$ on Y – a strong argument for the scale-bar/video measurement of d .

10.6 Random errors: tracking noise (sub-pixel), frame-timing jitter, release repeatability, d placement – half-range over 3–5 trials; bin-centring $\pm 1/(2T)$ unless interpolated.

11 Further methodology (≈ 200 words)

Numbered modified procedure (INTERNATIONAL §11.2 style):

1. Measure f_a, f_b of each spring alone (with its own magnet).
2. Track **both** magnets; form $\eta = x_1 + x_2$ and $\xi = x_1 - x_2$ *before* the FFT. Each normal coordinate contains essentially one frequency (f_1 from η , f_2 from ξ), so the two-peak resolution problem disappears even at 45–60 mm. (Physically motivated, free, the single best improvement.)
3. Record ≥ 90 s at 30–60 fps, tripod, scale bar in frame; release $A \leq 3$ mm; plus one $A = 10$ mm run per d for the amplitude study. Archive the raw videos.
4. Re-analyse every record with rectangular / Hann / Blackman windows $+16\times$ zero-padding + parabolic peak interpolation (INTERNATIONAL Table 10 style), **and** with a time-domain two-damped-sinusoid least-squares fit (beats the $1/T$ limit; gives a statistical Δf).
5. Fit models to (d, f_1, f_2) by weighted non-linear least squares: (a) dipole $n = 5$; (b) free exponent n ; (c) finite-size disc force; (d) detuned springs. Compare residual patterns and χ^2/dof (as a *relative* figure of merit – half-range uncertainties are not true σ ’s).

`analyse_tracks.py` performs steps 4–5 on a folder of Tracker exports.

12 Further data & discussion (≈320 words)

Table: gradient $\pm\Delta$ per method (rect / Hann / Blackman / two-tone fit / normal-coordinate FFT), as INTERNATIONAL Figs. 26–27. Graph: f_2 vs amplitude (segments) with extrapolation to zero amplitude. Graph: $\ln Y$ vs $\ln d$, old vs new data, with the point-dipole and finite-size theory curves. State which correction moved the gradient toward -5 and by how much (expected order: amplitude correction makes it *steeper*, i.e. further from your -3.75 ; resolution fix and finite-size model account for the rest). Be explicit about any residual discrepancy and its meaning (“dipole law valid for $d \gtrsim 4D$ ”).

13 Numerical simulation (≈250 words)

RK4 integration of the full equations

$$m\ddot{x}_1 = -kx_1 - [F(d+x_2-x_1) - F(d)], \quad m\ddot{x}_2 = -kx_2 + [F(d+x_2-x_1) - F(d)], \quad F = C_F r^{-4} \quad (11)$$

(optionally the finite-size F), with measured k, m, C_F . Uses: (i) $f_2(A, d) \rightarrow$ the amplitude correction of §12; (ii) synthetic $x(t)$ at your fps and T through the *same* FFT pipeline $\rightarrow$ the bias caused by the analysis alone (test at $T = 40$ s: window peak-picking gave -3.3 to -3.5 , the two-tone fit recovered the true gradient). Code $\rightarrow$ Appendix G.

14 Conclusion & extensions (≈220 words)

Answer the RQ in three sentences (f_1 constant within $\dots$; $Y \propto d^{-n}$ with $n = \dots \pm \dots$; dipole law valid for $d \geq \dots$). Name the dominant systematic error and what fixed it. Extensions: vary μ_m (magnet grade) at fixed d to test μ_m^2 ; attractive orientation (softening $\rightarrow$ instability at a critical d); three springs; eddy-current damping with a copper plate.

4 FFT error analysis – what to write

INTERNATIONAL needed *amplitudes* (SNR): energy-correction factors and baseline subtraction. You need *frequencies*: the relevant FFT physics is different – say so.

1. **Resolution.** $X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n / N}$, bin spacing $1/T$. Two tones are separable if $|f_2 - f_1| \gtrsim 1/T$ (rectangular) or $\gtrsim 2/T$ (Hann). From your data:

d/mm	$f_2 - f_1/\text{Hz}$	bins at $T = 20$ s	$T_{\min}$ rect/s	$T_{\min}$ Hann/s
21	0.240	4.8	4	8
25	0.141	2.8	7	14
28	0.097	1.9	10	21
31	0.074	1.5	14	27
35	0.039	0.8	26	51
40	0.009	0.2	111	222

2. **Leakage & windows.** Rectangular: narrowest main lobe, -13 dB side lobes; Hann: 4-bin main lobe, -31 dB; Blackman: 6-bin, -58 dB. Two peaks of similar height and no strong interferer $\Rightarrow$ rectangular (or Hann) is the right choice – the opposite of the usual “Hann by default” advice and unlike INTERNATIONAL’s case. No amplitude correction factors are needed (you read positions, not heights).
3. **Peak pulling.** Overlapping main lobes push the maxima apart or merge them: synthetic test at $T = 20$ s gave ± 2 – 7 mHz at 21–31 mm, ± 10 mHz at 35 mm, failure at 40 mm – quote your own numbers from `ee_simulations.py` Part A with your real T, f_s .
4. **Bin centring (scalloping).** $\pm 1/(2T)$ unless you zero-pad and interpolate (parabola through the three bins around the peak). Zero-padding interpolates the spectrum; it does *not* add resolution.

5. **Damping linewidth.** For $e^{-t/\tau} \cos 2\pi ft$ the *amplitude* spectrum has FWHM $\sqrt{3}/(\pi\tau) \approx 0.022$ Hz at $\tau = 25$ s (the power spectrum: $1/(\pi\tau) = 0.013$ Hz). Comparable to $f_2 - f_1$ at 35–40 mm, so beyond ~ 35 mm no spectral method on a single coordinate can separate the modes $\rightarrow$ normal-coordinate tracking and time-domain fitting.
6. **Sampling & pre-processing.** $f_s = \text{fps} \gg 2f$: no aliasing. Video frame timing jitters and frames drop $\rightarrow$ resample to a uniform grid (the script reports the jitter). *Detrend* (remove mean and linear drift): in pixel coordinates the DC offset is $\sim 50\times$ the oscillation amplitude, and with a rectangular window its leakage at ~ 16 bins (≈ -34 dB) is comparable to the signal peaks; the offset is present at every d (the lowest peak is always $f_1 \approx 0.8$ Hz).
7. **Propagation into the graph.** $\Delta(\ln Y) = 2(f_1\Delta f_1 + f_2\Delta f_2)/Y$ grows from 0.04 to 1.1 across your range for $\Delta f = 5$ mHz $\Rightarrow$ draw error bars in $\ln$ -space and weight the fit, else the noisy large- d points dominate. Horizontal bars $\Delta(\ln d) = \Delta d/d$ (4.8% at 21 mm for $\Delta d = 1$ mm) are comparable to the vertical ones at small d and enter any d^{-5} comparison as $5\Delta d/d$.

5 “Non-linearisation” – three levels, use all three in order

Level 1 – analysis (cheap, now).

Replace “take logs, then LINEST” by a weighted non-linear least-squares fit of $Y = Cd^{-n}$ (and $Y = Cd^{-5}$, n fixed) to the untransformed data. Advantages: no point discarded because $\ln(\leq 0)$ fails; heteroscedastic errors handled by weights; C and n with proper uncertainties.

Your 6 points: $n = 3.5 \pm 0.2$ (vs $\ln\text{--}\ln -3.75$) – report both, explain the difference.

Level 2 – physics (the systematic-error story).

Keep the full r^{-4} (or finite-size) force: perturbation formula (Appendix D) in the body, RK4 (§13) for exact numbers, amplitude-segment analysis + extrapolation to $A \rightarrow 0$ for the data.

Tabulate the correction at each d . Remember its sign: it makes the slope *steeper*.

Level 3 – model (finite size + detuning).

Fit models (c) and (d) of §11 step 5; present residual patterns and χ^2/dof (relative) as the evidence for which assumption failed. Expected signatures: finite size $\rightarrow$ concave-down residuals in the $\ln\text{--}\ln$ plot; detuning $\rightarrow f_1$ drift and a floor in Y at large d .

6 Systematic-error summary table (Table 7 template)

Effect	Type	Sign on Y / gradient	Size (your set-up)	Remedy
FFT resolution, pulling, linewidth	measurement	either; large- d points lost; net flattening	$\pm 2\text{--}10$ mHz (21–35 mm, $T = 20$ s); 40 mm lost	$T \geq 90$ s; η, ξ tracking; two-tone fit
Amplitude non-linearity (hardening)	model (b)	f_2 up at small d ; gradient <i>steeper</i>	+9.8% (21 mm, $A = 10$ mm); +1.1% (35 mm)	$A \leq 3$ mm; extrapolate $A \rightarrow 0$; RK4
Finite magnet size (softening)	model (a)	gradient <i>flatter</i>	$-4.9 \rightarrow -3.75$ for $D = 6 \rightarrow 20$ mm	finite-size force in model; $d \gtrsim 4D$
Non-identical springs	model (c)	f_1 rises at small d ; Y floor at large d	from f_a, f_b (e.g. 0.793/0.849 Hz)	measure f_a, f_b ; de-tuned fit
Tilt	model (f)	force slightly weaker	0.5–1.5%	small A ; one-line estimate
Damping	model (e)	shift negligible; width limits 10.1	$\frac{1}{2}(1/4\pi f\tau)^2$; 0.02 Hz	—
Δd	random / systematic	$5\Delta d/d$ on Y	12% at 21 mm for 0.5 mm	scale bar in video

7 Lab / data checklist

- ☐ Magnets: D, t, M , grade (B_r) – for §10.3 and the theoretical C .
- ☐ Springs: L, b, h, μ ; static k ; f_a, f_b with own magnet mounted (primary calibration).
- ☐ Balance measurement of $F(r)$, 4–6 values of r (force law + μ_m).
- ☐ Re-run $d = 21 \dots 60$ mm, both magnets tracked, $T \geq 90$ s, $A = 3$ mm (+ one $A = 10$ mm run each); files `d21_trial1.csv...`; archive raw videos.
- ☐ Run `analyse_tracks.py` on the folder (`--out results`) → per-file diagnostics, `summary.csv`, gradient per method; then repeat on η, ξ .
- ☐ Magnitude sanity check: $\gamma = 2\pi^2 mY$ from the data against $\gamma = 6\mu_0\mu_m^2/(\pi d^5)$ from μ_m . If they differ by more than $\sim 2\times$, revisit coaxiality and the definition of d before writing §8.4.
- ☐ Re-derive the $f_2 - f_1$ requirement with your real T ; if the original records were 20–40 s, say in §7 that 35 mm was marginal and 40 mm unresolvable.

Writing order. Collect data → §5, §6 → §4 → §7–8 → §9–13 → §1–3 and §14 last (contents page last of all). Record the pivots as they happen (period method → FFT; FFT limit → normal coordinates; linear → non-linear model) – they are the material for the reflective statement.

8 Assessment-criteria map (first assessment May 2027, 30 marks)

Criterion (marks)	Where the plan earns it
A Framework for the essay (6)	Focused RQ (§1.3); variables incl. FFT settings and the definition of d (§4); method choices justified, attempt 1 → 2 (§6.3); structure and presentation conventions (numbered figures/tables, appendices) – presentation now sits in A.
B Knowledge and understanding (6)	Full derivation chain (§5); DFT/FFT theory (§6.1); correct vocabulary (normal modes, coupling, spectral leakage, resolution, hardening/softening).
C Analysis and line of argument (6)	Quantitative hypothesis; processed data with propagated uncertainties on both axes (§8); theory–experiment comparison (§8.4); the argument runs prediction → measurement → two error families → resolved conclusion.
D Discussion and evaluation (8)	Every systematic effect with correct <i>sign</i> , size and remedy (§10, Table 7); improvements quantified (§11–12); simulation as independent check (§13); honest residual discrepancy and validity range (§14).
E Reflection (4)	The 500-word reflective statement (RPF), not the essay body: document the pivots above as they happen.

Sessions before May 2027 use the legacy 34-mark scheme (Critical thinking 12, Presentation 4, Engagement 6); confirm with your supervisor.

9 Figures and tables (final list for the essay)

Figures: 1 set-up sketch (done) · 2 mode sketch · 3 set-up photo · 4 $x(t)$ with beats · 5 $x(t)$ without visible envelope · 6 FFT with two peaks · 7 F vs x (k) · 8 F vs r on the balance (μ_m) · 9 $\ln Y$ vs $\ln d$ · 10 Y vs d^{-5} · 11 f_1 vs d · 12 synthetic-spectrum resolution figure · 13 f_2 vs amplitude · 14 gradient by method · 15 old vs new data with dipole and finite-size theory · 16 RK4 vs linearised prediction.

Tables: 1 constants · 2 k data · 3/4 raw f_1, f_2 · 5 processed · 6 theory vs experiment · 7 systematic-error summary · 8 further raw data (appendix) · 9 gradient/ C per method and model · 10 simulated vs measured f_2 shift.

10 Supporting figures already generated

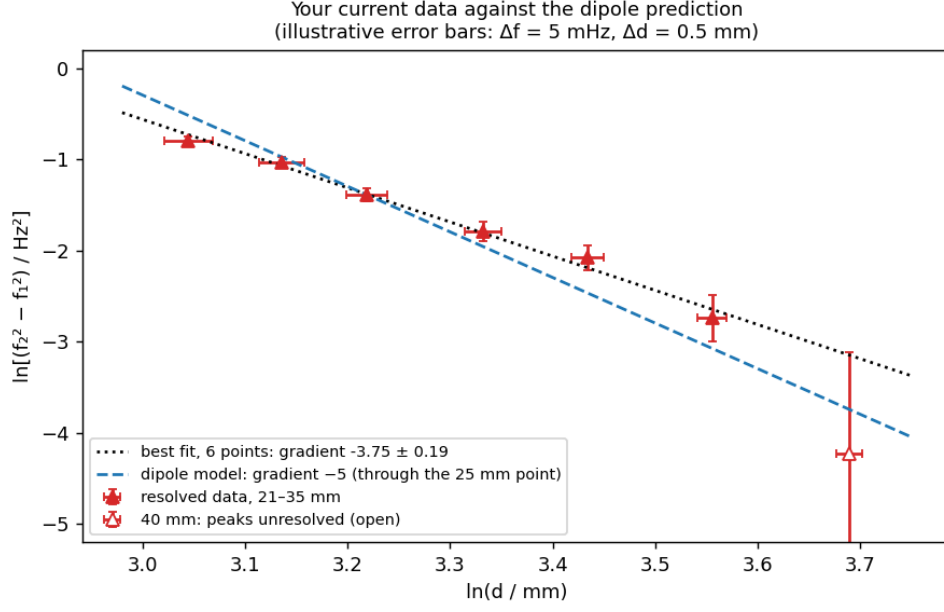


Figure 1: Your current data against the dipole prediction. The 40 mm point (open) sits far below the resolved trend, consistent with the resolution limit of Fig. 2, not with new physics. Error bars are illustrative ($\Delta f = 5$ mHz, $\Delta d = 0.5$ mm) and show why the $\ln$ -transform must be weighted. Use in §8.2/§9.

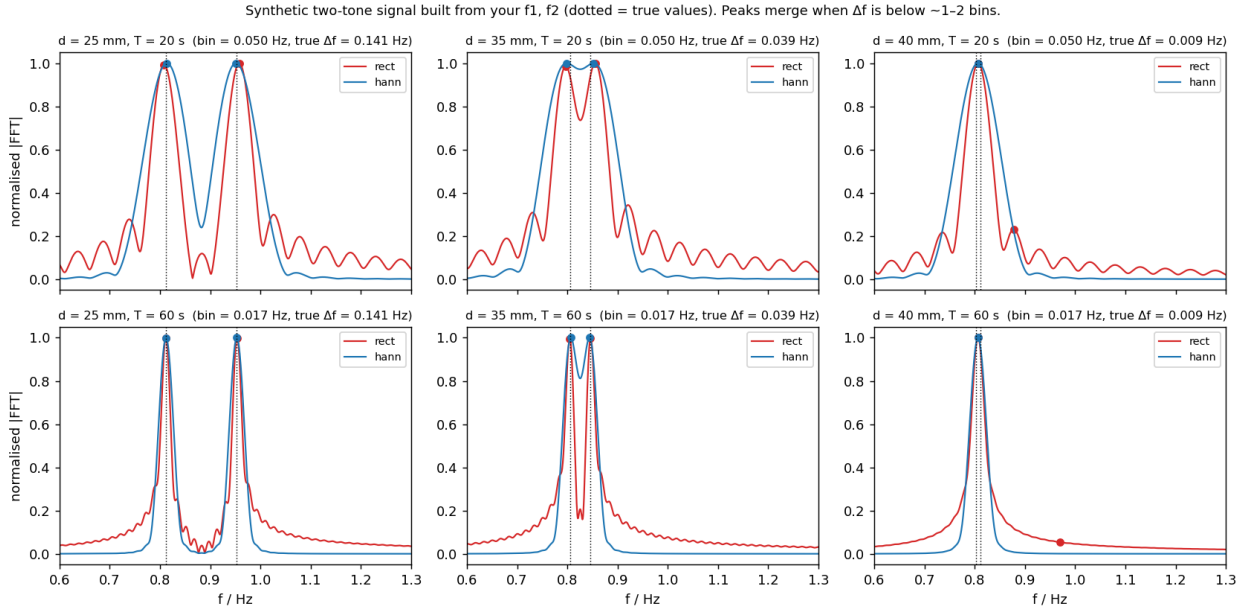


Figure 2: Synthetic two-tone spectra built from your f_1, f_2 (dotted = true values) for $T = 20$ and 60 s. Peaks merge when $f_2 - f_1$ falls below ~ 1 –2 bins. Use in §10.1.

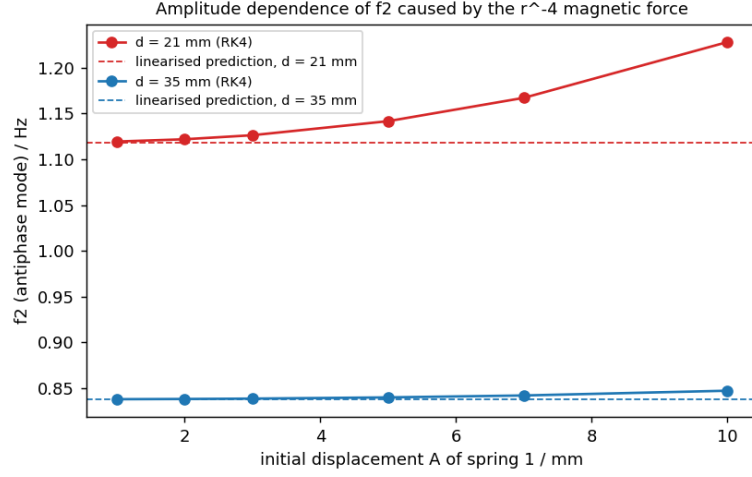


Figure 3: RK4 simulation: f_2 rises with release amplitude (hardening) relative to the linearised prediction. Use in §10.2/§13.

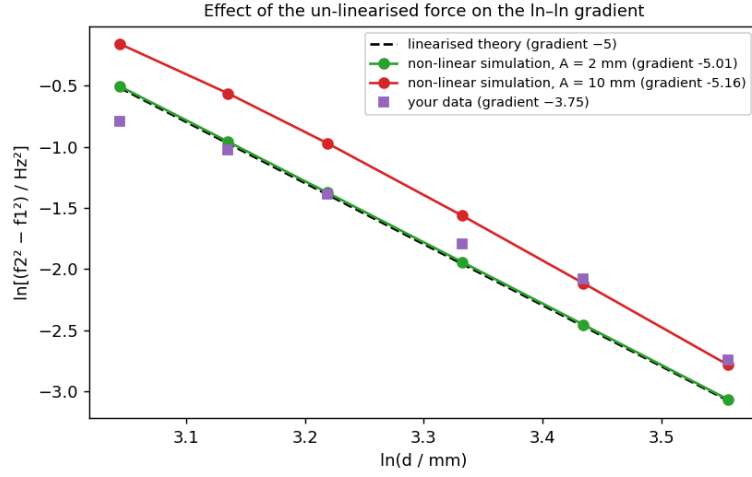


Figure 4: Effect of the un-linearised force on the ln-ln gradient: it becomes *steeper* (-5.16 at $A = 10$ mm), i.e. the amplitude effect moves away from your measured -3.75 . Use in §10.2.

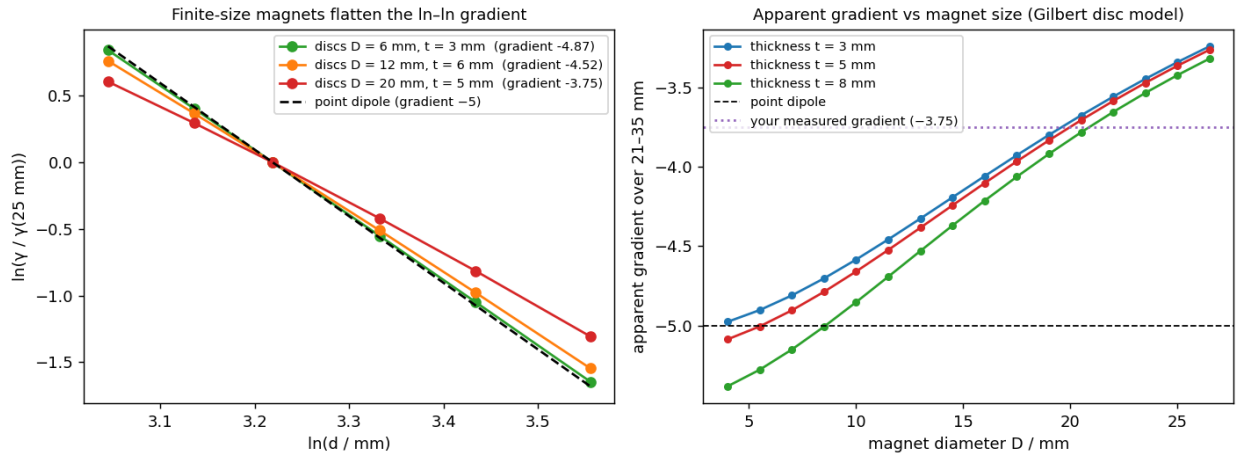


Figure 5: Finite-size (charged-disc) magnet model: the apparent gradient over 21–35 mm flattens as the magnet diameter grows. Use in §10.3.

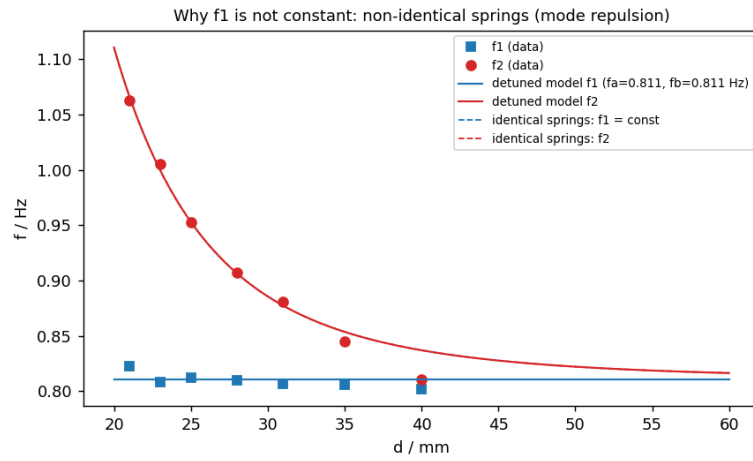


Figure 6: Non-identical springs: f_1 rises toward $\bar{f}$ as the coupling grows while f_2 keeps increasing. Use in §10.4.

11 Appendices to write

- A** Cantilever stiffness $3EI/L^3$, effective mass $\frac{33}{140}\mu L$, FEM check of lumped vs beam ($< 1\%$); dynamic calibration $f = \frac{1}{2\pi}\sqrt{k/m}$ and why m still needs the static k .
- B** Dipole field $\rightarrow$ coaxial dipole force $F = 3\mu_0\mu_m^2/(2\pi r^4) \rightarrow \gamma = 6\mu_0\mu_m^2/(\pi d^5)$; tilt factor $3\cos\theta_1\cos\theta_2 - \cos(\theta_1 - \theta_2)$.
- C** Detuned normal modes and the limits $\kappa \rightarrow 0$, $\kappa \gg \Delta$.
- D** Expansion of $F(d - \xi)$, α, β , Landau–Lifshitz shift, comparison with RK4.
- E** Finite-size (charged-disc) force and apparent-gradient table.
- F** Uncertainty propagation: ΔY , $\Delta \ln Y$, $\Delta \ln d$, Δd^{-5} , LINEST worst lines.
- G** Python: `analyse_tracks.py`, `ee_simulations.py`, `finite_size_magnets.py`.
- H** Further raw data (all windows/methods), as INTERNATIONAL Appendix A.

12 Ready-made numbers

Numbers to regenerate with your own inputs

Assumed in the scripts (marked # ASSUMED): f_s , τ , magnet size and mass. Regenerate every number below before it enters the essay.

- ln–ln fit, 6 points: gradient -3.75 ± 0.19 , intercept 10.68; weighted ($\Delta f = 5$ mHz): -3.48 ± 0.24 ; with the 40 mm point included: -4.9 (spurious). Free power law on raw Y : $n = 3.5 \pm 0.2$.
- Non-linear force, $A = 10$ mm: $f_2 + 9.8\%$ (21 mm), $+1.1\%$ (35 mm); gradient $-5.00 \rightarrow -5.16$. $A = 2$ mm: $+0.3\%$, $+0.03\%$.
- Finite-size discs, apparent gradient over 21–35 mm: $D = 6$ mm -4.87 ; 10 mm -4.6 ; 12 mm -4.5 ; 15 mm -4.2 ; 20 mm -3.75 ; 25 mm -3.4 ($t = 3$ –6 mm changes these by < 0.1).
- Lumped vs beam: Y differs by 0.03% ($M/\mu L = 3.3$), 0.4% (0.8), 1% (0.4).
- Detuned model with $n = 5$: $f_a = 0.793$, $f_b = 0.849$ Hz reproduces f_1 : $0.798 \rightarrow 0.819$ Hz.
- Tilt: $\theta = 0.10$ rad at $A = 10$ mm, $L = 150$ mm; interaction weaker by 0.5 – 1.5% .
- Synthetic tracks, $T = 40$ s: window peak-picking -3.3 to -3.5 ; two-tone fit -4.9 (true -4.9).

13 Pitfalls (examiner’s eye)

- Do not present the 40 mm point as data; present the unresolved range as a measured limit of the method.
- Do not claim “ f_1 constant” when it drifts 2.6% ; explain it (§10.4).
- Get the *sign* of every systematic effect right and say which way it moves the gradient; the amplitude effect steepens, finite size and detuning flatten.
- Do not copy IYPT slide figures or equations; cite the problem, derive the lumped model.
- Every graph: error bars on both axes, best/worst lines or LINEST uncertainty, units, gradient with uncertainty, theory line overlaid.
- Word count: equations, tables, captions, appendices excluded – move derivations there.